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Futures price prediction of agricultural products based on machine learning

Hailei Zhao[1]

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Abstract
Agricultural product futures are crucial to economic development, and the prediction of agricultural product futures prices has an important impact on the stability of the market economy. In order to improve the accuracy of agricultural product futures price prediction, based on machine learning algorithms, this study mainly uses machine learning methods to predict futures prices based on the analysis of fundamental factors affecting agricultural product futures prices. Moreover, in this study, wavelet analysis method is used to smooth the data and then build a model to process the hierarchical information after signal decomposition. In addition, this study conducts model validity studies through cases to draw comparative statistical diagrams to analyze the accuracy of model prediction data. The research shows that the model proposed in this paper has certain effects and can provide theoretical reference for subsequent related research.

Keywords Machine learning · Agricultural products · Futures · Price prediction

1 Introduction

At this stage, China's futures market as a whole is in a state of steady development, and it has relatively complete market operation standards and strong market supervision and management, and the sports committee in the international futures market is also gradually improving [1]. As a large agricultural country recognized by the world, China's agricultural products remain the most important part of the Chinese economy. Agricultural product futures are also one of the important components in the agricultural product industry chain, and they provide better trading tools for spot traders and investors by cooperating with the spot market. In the early days of conducting futures market trials in China, agricultural commodity futures are representative of agricultural product varieties. As the market has grown and improved over the years, agricultural commodity futures have become the most important agricultural product variety on the China Futures Exchange [2].

The futures market has a large leverage, so it has investment risks. It is a sharp double-edged sword. The futures market used as an investment tool can help investors transfer the risk of market price changes if it is used properly. If it is not used, it will bring investors multiple economic losses. Because of the above reasons, in the field of commodity futures research, how to effectively predict the futures price trend so as to optimize investment strategies to avoid price risks and improve investment returns has been a hot research direction. Agricultural product futures occupy a large trading volume in China's futures market, and price fluctuations have a wide range of influences. Therefore, the research and forecast of their price trends have become increasingly important. Statistical models have always had obvious advantages for forecasting research, especially in financial markets where the types of influencing factors are complex and numerous, and the models hidden in the data are difficult to show easily [3]. The commodity futures market, like other financial markets, is affected by complex factors such as national policies, macroeconomics, industrial economy, and investor psychology. In addition, the commodity futures market will also be affected by various factors upstream and downstream of the commodity variety industry chain, so its changes are extremely complicated, and the cycle is

✉ Hailei Zhao
zhhl@jiangnan.edu.cn

[1] School of Business, Jiangnan University, Wuxi 214122, Jiangsu, China

disorderly. This is a huge challenge for forecasting price movements. For a hundred years, prediction methods for futures price trends have continued to develop, and many people have tried to use traditional regression analysis to make predictions [4]. With the development of computer technology, the level of mathematical statistics has also been improved accordingly. Therefore, artificial neural networks with powerful nonlinear computing capabilities have begun to appear and have been widely used. The most outstanding feature of neural networks is their ability to fit nonlinear functions. This feature is the main reason why neural networks have huge growth potential in the financial field. The research focus of this article is whether the machine learning model can be applied to the prediction and simulation of the commodity futures market, and whether the application is reasonable and superior.

2 Related work

The price prediction of financial commodities has always been an important direction for scholars in various countries. The basic method is to analyze and predict time series through various models. For example, Mingyun Huo chose three A + H bank stocks and used time series analysis methods to model their daily logarithmic ARMA, GARCH, or EGARCH models. Then, by comparing their variance series, their risk levels were compared and analyzed [5]. Yingshuang Bai proposed a one-step forward static prediction method for stock price series by establishing an ARIMA model for stock prices and used this method to make short-term stock price predictions [6]. The literature [7] used the stock trading volume and its rise and fall as threshold variables, established a threshold autoregressive model and fitted it with actual data, and found that the model's fitting effect is better than ARCH, GARCH, CHARMA, and other models. The literature [8] et al. used the GARCH model and the BP neural network model for modeling based on different perspectives of the basic and technical analysis schools and compared the differences between the two models in stock price prediction. In the literature [9], in order to obtain a more reasonable stock investment opinion, a genetic algorithm was used for global optimization, and a neural network was used for local optimization. This model greatly improves the convergence speed of BP neural network while reducing the prediction error and uses the improved model to make stock prediction.

In the early stages of market growth, research on the futures market basically focused on the factor of price. The earliest proof that the price and trading volume of futures are related to the volume–price relationship research by Epps, Clark, Cornell, Karloff, and they proposed a positive correlation between price and trading volume. Later, David, Robert, Mahmoud, Andersen put forward and described the dynamic characteristics of the futures market. This research result directly affected many literatures on futures prediction by neural networks [10]. The literature [11] carried out an empirical study on the relationship between the futures spot price of copper and the commodity. The literature [12] researched the volume–price relationship in China's futures market and the calendar effect of price futures in China's futures market. In the field of price prediction, many researchers in China use traditional statistical methods to make price predictions on the futures market. The literature [13] studied whether AR model has good application in futures price prediction. The literature [14] used ARMA model to make a time series to perform local stabilization and predict futures prices based on this. Later, many researchers and scholars used the ideas of EWMA and GARCH models to make futures price prediction models. The use of neural networks in China's futures volume and price research is far less than foreign research, and even far less than the application of neural network methods in the stock market. Therefore, this is where China's futures price research and prediction are lacking. Foreign countries already have a lot of experience to learn from, so we can try to use data mining methods such as neural networks to better study futures prices.

The literature [15] used neural network to predict S & P and gold futures prices in the futures market. The literature [16] predicted the exchange rate of the yen based on a feedforward network model. The literature [17] used neural network model method to predict the price of S & P 500 index futures volatility. The literature [18] and others built a wavelet-based oil price forecasting model to classify and merge the disordered data in the original data.

The literature [19] integrated multiple prediction methods to finally construct a more accurate forecast model of exchange rate, which was trained by a single-layer neural network. The literature [20] combined HMM, artificial neural network, and genetic algorithm to propose a fusion model to predict the behavior of financial markets and predict the price of the next day. The literature [21] compared the results of gold futures price prediction based on ANN and ARIMA models, and confirmed that ANN was significantly better than ARIMA, and analyzed the reasons for the results.

3 Related methods of agricultural product futures price prediction models

3.1 Principal component analysis

Principal component analysis (PCA) is a more important method for reducing the dimension of data in statistics.

When using this method to perform dimension reduction on multidimensional data, the basic steps are as follows:

(1) Standardized extraction of raw data indicators. For the P-dimensional random vector g in the original data of each sample, the corresponding normalized transformation is performed according to Formula (1) to obtain a normalized matrix z.

$$z_{ij} = \frac{x_{ij} - \bar{x_j}}{\delta_j}, \quad i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, p \tag{1}$$

Among them, $\left|R - \lambda I_p\right| = 0, \delta_j = \sqrt{\frac{\sum_{i=1}^{n}\left(x_{ij} - \bar{x_j}\right)^2}{n}}$ and p are the number of variables in the sample data, and n is the corresponding data number of each variable.

(2) By solving the normalization matrix Z, the correlation coefficient matrix $R\left(r_{ij}\right)$ is obtained. The formula is shown as follows:

$$r_{ij} = \frac{\delta_{ij}}{\sqrt{\delta_{ij}} \times \sqrt{\delta_{jj}}}, \delta_{ij} = \frac{\sum_{k=1}^{n}\left(x_{kj} - \bar{x_i}\right)\left(x_{kj} - \bar{x_j}\right)}{n-1}. \tag{2}$$

(3) By solving the characteristic equation $\left|R - \lambda I_p\right| = 0$ of the correlation matrix $R\left(r_{ij}\right)$, the eigenvalues and eigenvectors are obtained. We can obtain P eigenvalues $\lambda_1 \gg \lambda_2 \gg \cdots \gg \lambda_p$ and their corresponding eigenvectors β_i. Among them, if λ_i represents the variance of the i-th principal component y_i, then the ratio of λ_i to the total variance of all principal component factors is called the variance contribution ratio $F = \sum_{i=1}^{K} \alpha(y_i) y_i$ of the principal component factor y_i. The formula is shown as follows:

$$\alpha(y_i) = \frac{\lambda_i}{\sum_{i=1}^{p} \lambda_i}. \tag{3}$$

(4) The specific number of principal components was determined. The general selection criterion is to select the first K principal component factors. The sum $\alpha(y) = \sum_{i=1}^{K} \alpha(y_i)$ of the variance contribution ratios of these K factors exceeds a given value. This value is usually set to 90% (different requirements are different).

(5) The obtained principal component factors were comprehensively evaluated. The final evaluation value F can be obtained by taking the respective variance contribution rates $\alpha(y_i)$ of the first K principal component factors as weights and performing linear weighted summation on the K principal components. The formula is shown as follows:

$$F = \sum_{i=1}^{K} \alpha(y_i) y_i. \tag{4}$$

3.2 Independent component analysis

Independent component analysis (ICA) is a relatively new data processing technology, which refers to a technology of separating source signals from a linear mixed signal of multiple source signals.

(1) Principles of the ICA model

The basic principle of ICA originates from the classic "cocktail party." If it is assumed that there are persons at the banquet who can speak at the same time and n sound receivers are placed in some corners of the room to record sound, then a set of data $\left\{x^{(i)}\left(x_1^i, x_2^i, \ldots, x_n^i\right); i = 1, 2, \ldots, m\right\}$ is collected from the microphones after the banquet. Among them, i represents the time sequence of sampling. It can be seen that a total of m sets of sampling data are collected, and each set of sampling information is n-dimensional. Moreover, our goal is only to distinguish the source of each person's speech from these m groups of samples.

By refining the above cocktail problem, we can get: There are n signal sources,$S(S_1, S_2, \ldots, S_n)^T, S \in R^n$.. Among them, each dimension is a person's voice and is relatively independent. A is an unknown mixing matrix used to combine the superimposed signal S. Then, the following formula exists:

$$X = AS. \tag{5}$$

In the above formula, X is a matrix. Among them, each component of each column vector $X^{(i)}, X^{(i)} = AS^{(i)}$ is linearly represented by $S^{(i)}$. Both A and S are unknown, and X is known, so we have to find a way to derive S based on X. This process is the precipitation of independent components, also known as blind signal separation.

If we assume $W = A^{-1}$, then $S^{(i)} = A^{-1}X^{(i)} = WX^{(i)}$. W is expressed as the following matrix:

$$W = \begin{bmatrix} -w_1^T- \\ -w_2^T- \\ -\vdots- \\ -w_N^T- \end{bmatrix}. \tag{6}$$

Among them, $w_i \in R^n$. Essentially, writing w_t as a row vector, we get the following formula:

$$S_j^{(i)} = W_j^T X^{(i)}. \tag{7}$$

(2) Algorithm of ICA model

The ICA algorithm was developed by two scholars, Bell and Sejnowski. The study of the algorithm can be divided

into two categories: iterative estimation methods based on information theory and algebra methods based on statistics. The maximum likelihood algorithm of the former is used in the research of this paper. The principle of the maximum likelihood estimation algorithm is briefly introduced below.

If it is assumed that each S_i has a probability density p_s, then the joint distribution of the source signals at a given time is shown by the following formula:

$$p_{(s)} = \prod_{i=1}^{n} p_s(s_i). \tag{8}$$

Formula (8) represents a hypothetical premise that each person's sound signal is independent. After $p_{(s)}$ is obtained, we can obtain $p_{(x)}$ as shown in Formula (9):

$$p_{(x)} = p_s(W_x)|W| = |W| \prod_{i=1}^{n} p_s\left(w_i^T x\right). \tag{9}$$

The left side of Formula (9) is the probability of each sampled signal X (n-dimensional vector), and the right side is $|W|$ times the product of the probability of each source signal. Therefore, we need to know $p_s(s_i)$. We first select a probability density function to assign to S, and the density function cannot be Gaussian distributed. From the knowledge of probability theory, we know that the density function $P(x)$ can be obtained after the derivative of the cumulative distribution function $F(x)$.$F(x)$ needs to satisfy two properties: monotonically increasing and range in $[0, 1]$. The domain of the Sigmoid function is $[0, 1]$, and it is a slowly increasing function, so it fits. Therefore, in view of the above reasoning, it is assumed that the cumulative distribution function of S conforms to the Sigmoid function. Its function expression is shown in Formula (10):

$$g(s) = \frac{1}{1 + e^{-s}}. \tag{10}$$

The derivative result is shown in Formula (11):

$$p_s(s) = g'(s) = \frac{e^s}{(1 + e^s)^2}. \tag{11}$$

The above formula is the density function of S, and S is a real number. In the above formula, $p_s(s)$ is a symmetric function. Therefore, $E[S] = 0$.Then, $E[X] = E[S] = 0$, that is, the mean of X is also 0.

Since $p_s(s)$ is solved, only W is unknown. At a given sampling training sample $\left\{x^{(i)}\left(x_1^i, x_2^i, \ldots, x_n^i\right);\ i = 1, 2, \ldots, m\right\}$, the likelihood estimation of the logarithm of the sample is shown in Formula (12), which is obtained using the probability density function of X obtained previously:

$$l(W) = \sum_{j=1}^{n} \left\{\log g'\left(w_j^T x^{(i)}\right) + \log|W|\right\}. \tag{12}$$

In Formula (12), $P\left(x^{(i)}\right)$ is shown in braces. This is the derivative of the determinant. The derivative is shown in Formula (13):

$$\nabla_w |W| = |W|\left(W^{-1}\right)^T. \tag{13}$$

Then, the finally obtained derivative formula result is shown in Formula (14):

$$W = W + \alpha \left(\begin{bmatrix} 1 - 2g\left(w_1^T x^{(i)}\right) \\ 1 - 2g\left(w_2^T x^{(i)}\right) \\ \vdots \\ 1 - 2g\left(w_n^T x^{(i)}\right) \end{bmatrix} \right) x^{(i)T} + \left(W^T\right)^{-1}. \tag{14}$$

Among them, α is the artificially set gradient rising rate. When iteratively finds W, then $S^{(i)} = Wx^{(i)}$ can be obtained to restore the original signal.

3.3 Granulation of fuzzy information

For a given time series $X = (x_1, x_2, \ldots, x_N)$, the entire time sequence X is regarded as a window for blurring. A fuzzy concept G (fuzzy set with X as the domain of discussion) that can reasonably describe X is constructed, and then, fuzzy particles are determined, which can be expressed as Formula (15):

$$P = A(x), x \in X. \tag{15}$$

There are four types of fuzzy particles that are commonly used, namely triangular, Gaussian, ladder, and parabolic. Because this article is to predict the fluctuation range of agricultural futures prices, it is necessary to know the maximum, average, and minimum values of the window. Therefore, triangular fuzzy particles are used in this study, and their membership functions are shown in Formula (16):

$$A(x, a, m, b) = \begin{cases} 0, x \prec a, x \succ b \\ \dfrac{b - x}{b - m}, m \prec x \leq b \\ \dfrac{x - a}{m - a}, a \leq x \leq m \end{cases}. \tag{16}$$

4 Introduction to the principles of the predictive model method

4.1 Support vector machine classification method

For the training set (x_i, y_i), the nonlinear support vector machine needs to solve the following optimization problem, which can be solved by Formula (17):

$$\min_{\omega,\omega_0} J_{svm}(\omega,\omega_0) = \frac{1}{2}\|\omega\|^2 + c\sum_{i=1}^{n}\delta_i. \tag{17}$$

Among them, the constraint condition is shown in Formula (18):

$$\begin{aligned} &Z_i\left(\omega^T x_i + \omega_0\right) \geq 1 - \delta_i, i = 1,2,\ldots,n \\ &\delta_i \geq 0, i = 1,2,\ldots,n. \end{aligned} \tag{18}$$

In addition, we need to introduce the Lagrange coefficients α and β to obtain the Lagrangian function as shown in Formula (19):

$$\begin{aligned} L(\omega,\omega_0,\delta_i,\alpha,\beta) &= \frac{1}{2}\|\omega\|^2 + C\sum_{i=1}^{n}\delta_i \\ &- \sum_{i=1}^{n}\beta_i\delta_i - \sum_{i=1}^{n}\alpha_i\left[Z_i\left(\omega^T x_i + \omega_0\right) - 1 + \delta_i\right]. \end{aligned} \tag{19}$$

The partial derivatives of ω and ω_0 in Formula (19) are obtained and are brought into Formula (17). By simplification, the dual problem of Formula (20) can be obtained:

$$\min_{\alpha} L(\alpha) = \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j z_i z_j K\left(x_i x_j\right). \tag{20}$$

The corresponding constraint is shown in Formula (21):

$$\begin{aligned} &\sum_{i=1}^{n}\alpha_i z_i = 0 \\ &C \geq \alpha_i \geq 0, I = 1,2,\ldots,n. \end{aligned} \tag{21}$$

In Formulas (20) and (21), z is the number of training samples, C is a penalty factor, and $K\left(x_i x_j\right)$ is a kernel function.

The nonlinear SVM discriminant function in the input space only needs to use the kernel function to calculate the inner product $K(x,x_i)$ of the test sample X and the training sample g in the feature space:

$$g(x) = \sum_{i=1}^{n}\alpha_i z_i K(x,x_i) + \omega_0. \tag{22}$$

Support vector machine theory cleverly solves the problem of mapping low-dimensional input space to high-dimensional feature space by introducing a kernel function, so we only need to produce $g(x)$ in Formula (22) to achieve our desired result.

4.2 Support vector machine regression

The given training set samples are set to

$T = \{(x_k,y_k)|k = 1,2,\cdots,n\}, x_k \in R^n, y_k \in R.x_k$ is the input value and y_k is the output value. According to the principle of minimizing structural risk, the optimization problem is transformed into finding the minimum function $f(x)$:

$$\begin{aligned} &\text{Min } J(\omega,e,b) = \frac{1}{2}\omega^T\omega + \frac{1}{2}\gamma\sum_{i=1}^{n}e_i^2 \\ &\text{s.t. } y_i = \omega^T\varphi(x_i) + b + e_i, i = 1,2,\ldots,n \\ &\varepsilon_i \geq 0, i = 1,2,\ldots,l. \end{aligned} \tag{23}$$

In Formula (23), ω is a weight vector and γ is a penalty parameter, which indicates the degree of punishment for experimental samples that exceed the error e, b is an offset, $\varphi(x_i)$ is a kernel space mapping function, and $J(\omega,e,b)$ is a loss function.

The expressions of α and b obtained by the iterative algorithm are shown in Formula (24):

$$\begin{aligned} &\alpha = (y - bl_v)\beta^{-1} \\ &b = \left(l_v^T\beta^{-1}l_v^T\right)^{-1}l_v^T\beta^{-1}y. \end{aligned} \tag{24}$$

In Formula (24),

$$\begin{aligned} &\beta = ZZ^T + C^{-1}I, Z = \left[\varphi(x_1)^T;\ldots;\varphi(x_n)^T\right], \\ &y = [y_1,y_2,\ldots,y_n], l_v = [l_1;l_2;l_n], \\ &\alpha = [\alpha_1,\alpha_2,\ldots,\alpha_n]. \end{aligned}$$

According to the Mercer condition, the mapping function $\varphi(x)$ and the kernel function $K(x,x_i)$ have the following relationship:

$$K(x,x_i) = \varphi(x)^T\varphi(x_i). \tag{25}$$

Then, the fitting function of the support vector machine can be expressed as Formula (26):

$$f(x) = \sum_{i=1}^{n}\alpha_i K(x,x_i) + b. \tag{26}$$

5 Common technical indicators for agricultural product futures price prediction

5.1 Moving average

Principle: Moving average (MA), also known as moving average, cost line, is a widely used quantitative analysis tool in technical analysis and is an important technical analysis indicator. The basic principle is to use the principle of averages to eliminate irregular fluctuations in the series so that the average cost of the current commodity shows the true trend of prices to assist investors in predicting future price trends.

Calculation method: The calculation method is the average calculation of closing prices for several

consecutive days. The specific calculation formula is shown as follows:

$$M_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-1}. \tag{27}$$

Among them, n is the number of data taken by the moving average. The moving average is a curve of the closing price moving average time series M_t calculated by Formula (27) and drawn on the original closing price curve. It is generally believed that a close above MA means that the futures price is bullish, otherwise the futures price is bearish. In addition to the simple moving average obtained by Formula (27), there are weighted and nonlinear weighted moving averages. The calculation methods of the two are relatively complicated, and the effect is not better than that of simple MA. They are not described in detail here.

5.2 Exponential smoothing moving average

Principle: The exponential smoothing moving average (MACD) indicator is constructed based on the principle of moving averages. This indicator is obtained by applying a double smoothing operation using the fast-moving line, which is a short-term moving average, and the slow-moving line, which is a long-term moving average, and the signs of their intersection and departure. The advantage is that it not only retains the effect of normal MA, but also removes the defect that the MA frequently sends false signals. Therefore, the MACD indicator has the characteristics of stability and stability while having the trend of moving averages. It is an effective technical analysis indicator used to study the timing of securities trading and predict the rise and fall of securities prices. This indicator is mainly used to study the trend of futures prices through the relationship between DIF (positive and negative difference), DEA (average of similarities and differences), and BAR (bar graph). Among them, DIF is the core and DEA is the auxiliary.

Calculation method: MACD indicator is divided into the following steps in calculation:

(1) Calculation of DIF. DIF is the core point, which is the difference between fast and slow smooth moving averages. Currently, the short-term and long-term moving averages on the 12th and 26th are generally taken.

Step 1: The smoothed moving average (EMA) of the 12th and 26th is calculated. The formula is shown as follows:

$$EMA_t = \alpha P_t + (1-\alpha)EMA_{t-1}. \tag{28}$$

In the formula, α is the smoothing coefficient, and the formula $\alpha = \frac{2}{n+1}$ is calculated. Generally, α is taken as 12 days and 26 days, that is, α_{12} and α_{26}. Then, the calculation formulas of $EMA_t(12)$ and $EMA_t(26)$ are shown as follows:

$$\begin{aligned} EMA_t(12) &= \frac{2}{12+1}P_t + \frac{11}{12+1}EMA_{t-1} \\ EMA_t(26) &= \frac{2}{26+1}P_t + \frac{25}{26+1}EM. \end{aligned} \tag{29}$$

The EMA value on the first day can be taken as the closing price on the first day.

Step 2: The DIF value is calculated. The calculation formula is shown as follows:

$$DIF = EMA_t(12) - EMA_t(26). \tag{30}$$

(2) DEA is calculated. DEA is the moving average of DIF, and its calculation method is the same as the moving average calculation method introduced earlier. It just replaces DIF with the closing price.

(3) BAR is calculated. The calculation of the histogram BAR is relatively simple, and its calculation formula is shown as follows:

$$BAR = 2 \times (DIF - DEA). \tag{31}$$

Theoretically, in a continuous upward trend, the 12-day EMA line is above the 26-day EMA line, and the positive deviation value $(+DIF)$ between them has become a little larger. Conversely, in the continuous downward trend, the dispersion value may become negative $(-DIF)$ and gradually increase. When the market conditions of the securities improve, the difference between the positive and negative dispersion values will become smaller. The indicator MACD analyzes and judges futures trading signals based on the crossover principle of fast- and slow-moving lines. In addition, the degree of contraction of the BAR bars can also be used to execute buying and selling decisions.

5.3 Relative strength index

Principle: The economic theoretical basis of the relative strength index (RSI) is that the market supply and demand conditions determine the direction of price changes, and price changes in turn affect market demand. The specific principle is to analyze the strength and weakness of buyers and sellers by comparing the average decline of the closing price during the base period, so as to study the price trend of the futures. Generally, people use it in conjunction with MA in specific practical operations to improve the accuracy of price trend prediction.

Calculation method: The RSI calculation method uses the daily fluctuation of the closing price within a certain period of time (N days) to measure the comparison of the long and short power in these days. The RSI considers the

sum of the daily closing prices in the N days (that is, the closing price of the day is higher than the closing price of the previous day) as the total strength of the buyer and the decline in closing prices in the N days (that is, the closing price of the day is lower than the closing price of the previous day) as the total strength of the seller. The specific calculation formula is shown as follows:

$$RSI = 100 - \frac{100}{1 + RS}. \tag{32}$$

RS in Eq. (32) can be obtained from the following Eq. (33):

$$RS = \frac{\bar{P}_{\text{rise}}}{\bar{P}_{\text{fall}}}. \tag{33}$$

In the formula, $\bar{P}_{\text{rise}}$ represents the average value of the sum of the increase in closing prices during the base period, and $\bar{P}_{\text{fall}}$ represents the average value of the sum of the decrease in closing prices during the base period.

It is known from Formula (32) that the value of RSI is between 0 and 100. If the daily calculation period is used, traders usually use RSI with fewer days for short-term practical operations. At present, the benchmark periods chosen by the RSI for Shanghai and Shenzhen are 6th and 12th.

5.4 Stochastic

Principle: The stochastic indicator (KDJ) is a reliable and sensitive technical indicator currently used in the futures market. This indicator combines the statistical principle with the theory of MA and the theory of RSI's supply and demand balance point to form a more accurate buying and selling signal. It is calculated on the basis of the highest price, the lowest price, and the closing price of the futures product to obtain the K value, D value, and J value. Then, a point on the coordinates of the indicator is formed. By connecting countless such points, a complete KDJ indicator can be formed that can reflect the trend of price fluctuations.

Calculation method: The calculation of KDJ is more complicated than other indicators. Before the calculation, the RSV value of the period (n days, n weeks, etc.) must be calculated, that is, the Raw Stochastic Value, and then the K value, D value, J value, etc. can be calculated. Among them, RSV is the position ratio of the closing price on the last trading day between the highest price and the lowest price of the previous paragraph. The calculation formula is shown in Eq. (34), which takes the calculation of the daily KDJ value as an example:

$$RSV = \frac{C_t - L_n}{H_t - L_n} \times 100\%. \tag{34}$$

In Formula (34), C_t is the closing price of the research day, L_n is the lowest price data in the last n days, and H_t is the highest price data in the n days. The RSV value always fluctuates between 1 and 100. After the RSV is calculated, the K and D values can be calculated next. The calculation formula is shown as follows:

$$\begin{aligned} K_t\% &= \frac{1}{3} \times RSV\% + \frac{2}{3} \times K_{t-1}\% \\ D_t\% &= \frac{1}{3} \times K_t\% + \frac{2}{3} \times D_{t-1}\%. \end{aligned} \tag{35}$$

In the formula, $RSV\%$ represents the RSV of the day, $K_{t-1}\%$ represents the K value of the previous day, $K_t\%$ represents the K value of the day, and $D_{t-1}\%$ represents the D value of the previous day. In addition, it should be noted that when the K and D values are first calculated, we assume $K_0 = 50, D_0 = 50$ and bring them into the calculation. The smoothing factors $1/3$ and $2/3$ in the formula are artificially selected. At present, in many securities analysis software, they have been set to $1/3$ and $2/3$, without artificially changing them. In addition, when using a random index. The calculation formula is shown as follows:

$$J_t\% = 3D_t\% - 2K_t\%. \tag{36}$$

In fact, $J_t\%$ is the deviation between $K_t\%$ and $D_t\%$, which is an auxiliary indicator. The initial KDJ indicators were only the K-line and the D-line. With the development of securities market analysis technology, the KD indicator has gradually evolved into a KDJ indicator, and the ability of the indicator to analyze the market has improved a lot.

5.5 BISA

Principle: BISA is also abbreviated as Y value, which is an index derived and separated from MA. BIAS actually reflects the technical indicators of how far the futures price deviates from the MA in the process of market fluctuations. Its theoretical basis is that whether the futures price is above or below the MA, as long as it is too far away from the MA, the centripetal effect of cost will pull the futures price back to the vicinity of the MA. Based on this principle, the best time to buy and sell can be determined.

Calculation method: The calculation formula of the BISA of N days is shown as follows:

$$Y = \frac{P - \bar{P}_N}{\bar{P}_N} \times 100\%. \tag{37}$$

In the formula, P is the closing price of the current day, and $\bar{P}_N$ is the average price of the closing prices in N days. N can be taken on the 6th, 12th, 24th, 72nd, and 200th days. Generally, the forecast of short-term securities price trends takes 6th and 12th.

Table 1 Technical indicators of agricultural futures data

MA (5)	K% (6)
MA (10)	D% (6)
RSI (6)	J% (6)
RSI (9)	PXY (12)
BIAS (6)	DIF
BIAS (12)	DEA
R%	BAR

5.6 William index

Principle: The Williams indicator (*R%*) is also a technical indicator used to predict the short-term movement of futures prices. The principle is to analyze the overbought and oversold situation of the futures market by analyzing the relationship between the highest price, the lowest price, and the closing price of the futures price within a certain period of time. Specifically, it mainly uses the oscillation point to show the market's overbought and oversold behavior, and then analyzes the power of both long and short positions, so as to propose effective signals to study the market's short-term and short-term behavior.

Calculation method: William indicator is a relative indicator, which indicates the relative position of the closing price of the day within the entire price range in the past period. By subtracting the closing price of the day from the highest price in this period and comparing the difference with the entire price range in this period, the William's indicator for the day is obtained. The specific calculation formula is shown as follows:

$$R\% = \frac{H_n - C}{H_n - L_n} \times 100. \tag{38}$$

In the formula, C is the closing price of the calculation day, L_n is the lowest price data in the period, and H_n is the highest price data in the N period. In the formula, the calculation time parameter of the N is usually 4 or 14. The calculated $R\%$ value ranges from 0 to 100. The closer it is to 0, the closer the futures price is to the lowest price in the past N days, and the closer it is to 100, the closer the futures price is to the highest price in the past N days.

5.7 Popularity index psychological line

Principle: The psychological line (PSY) is a quantitative measure that reflects people's attitude toward this market. This indicator is a technical indicator that studies the psychology or fact that investors tend to be buyers or sellers over a period of time. That is, by using the time when the market potential rises over a period of time and the ratio curve of that period of time, whether the market is going long or short is being judged.

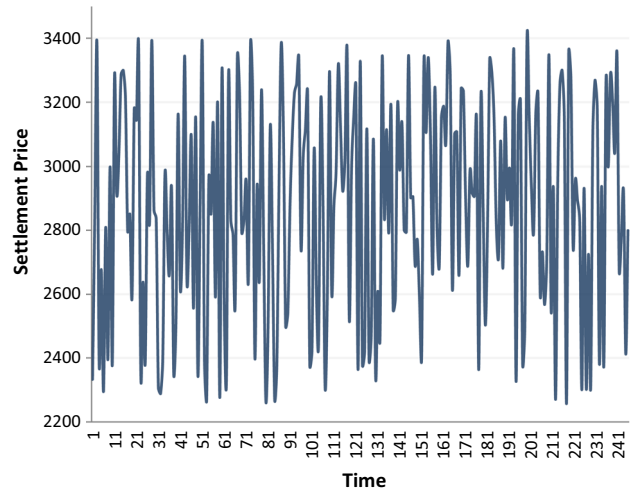

Fig. 1 Timing chart of futures settlement price

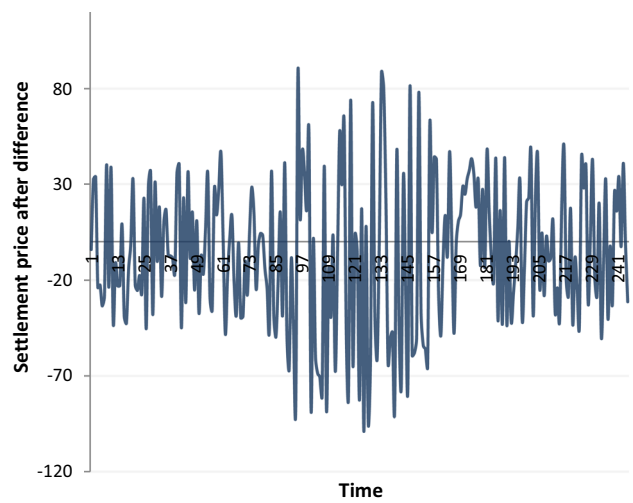

Fig. 2 Timing diagram after the difference of futures settlement price

The calculation method and formula are shown in Formula (39):

$$PSY = \frac{A}{N} \times 100\%. \tag{39}$$

Among them, A is the number of days when the closing price rises in N days. N is generally set to 12 days, and its maximum does not exceed 12. The maximum length of the weekly line does not exceed 26.

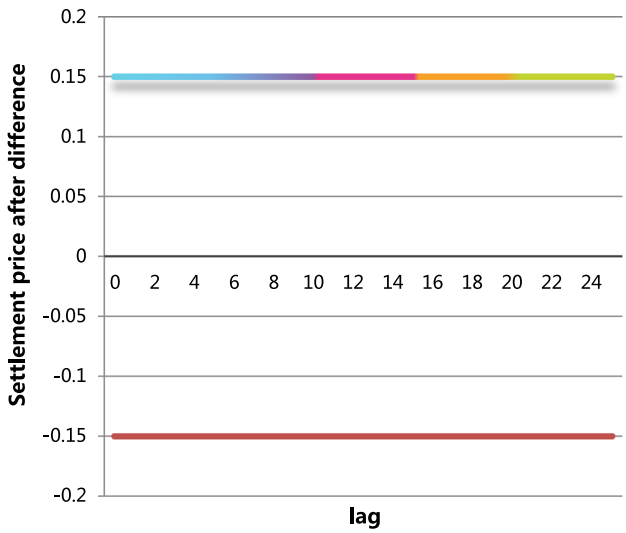

Fig. 3 Autocorrelation diagram after difference

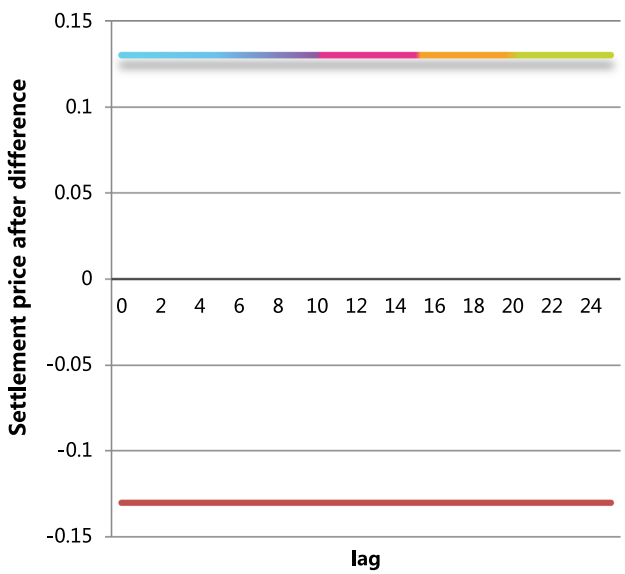

Fig. 4 Partial autocorrelation diagram after difference

6 Steps of constructing feature space of agricultural product futures price trend

6.1 Selection of technical index of agricultural product futures price feature data

The original historical transaction price data available on all trading days from January 4, 2009, to December 31, 2018, is a continuous series. The data available from the official website include a total of 10 basic indicators (opening price, closing price, highest price, lowest price, settlement price, trading volume, positions, gains, declines, transaction amount). After extracting some data features, 14 representative technical indicators can be calculated as

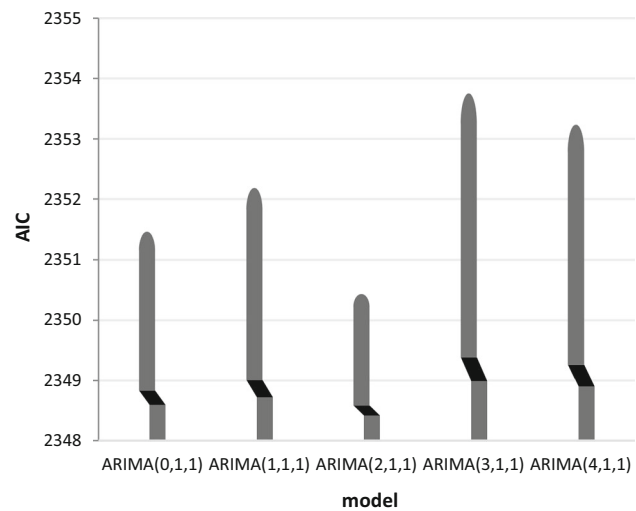

Fig. 5 Model comparison diagram

shown in Table 1. The 14 kinds of technical indicators are moving averages (MA5, MA10), relative strength indicators (RS16, RSI9), BIAS (BIAS6, BIAS12), Williams index (*R*%), random index (K% 6, *D*% 6, *J*% 6), psychological line (PXY12), positive and negative difference(DIF), mean and difference (DEA), and bar line BAR. The specific meanings of these indicators have been explained in the previous section, so they are not explained here.

6.2 Futures price feature extraction and establishment of trend prediction model

The detailed steps of extracting futures price characteristics and establishing a trend prediction model are as follows:

(1) Sample selection and data preprocessing. Firstly, in the historical price data of agricultural futures, the price and volume information features with trend segments are selected to extract 14 meaningful technical indicators, as shown in Table 1. The trend segment data are then normalized. The normalized formula is shown as follows:

$$y = \frac{(y_{\max} - y_{\min}) * (x - x_{\min})}{x_{\max} - x_{\min}} + y_{\min}. \tag{40}$$

In the formula, y represents the normalized agricultural futures price data, x represents the original agricultural futures price data to be standardized, and $x_{\max}$ and $x_{\min}$ represent the maximum and minimum values of the agricultural futures price data in the original sequence to be standardized. After normalizing the original data X, the range of the obtained data Y is $[y_{\min}, y_{\max}]$. Among them, we take $y_{\min} = 0, y_{\max} = 1$.

(2) Feature reduction and selection and extraction of model features: The PCA and ICA methods were used to reduce the dimensionality of 14 indicators, and the data of

Table 2 Model comparison table

	ARIMA (0,1,1)	ARIMA (1,1,1)	ARIMA (2,1,1)	ARIMA (3,1,1)	ARIMA (4,1,1)
AIC	2351.4618	2352.189	2350.4316	2353.7545	2353.2394
Significance test	Significantly	Significantly	Significantly	Not obvious	Not obvious

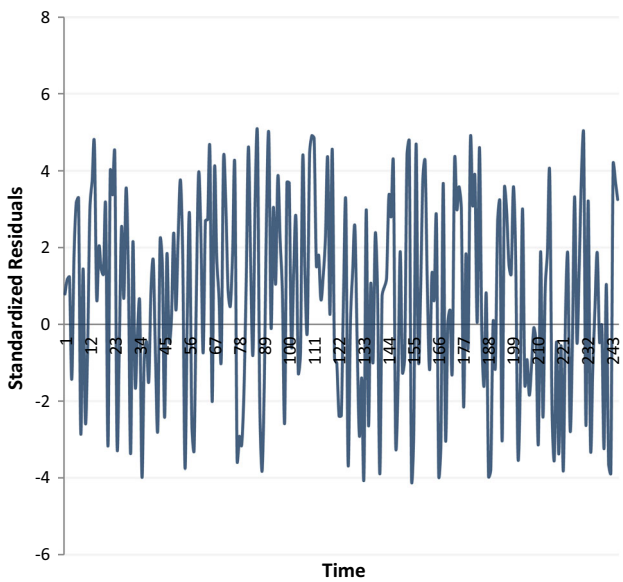

Fig. 6 Standardized residuals

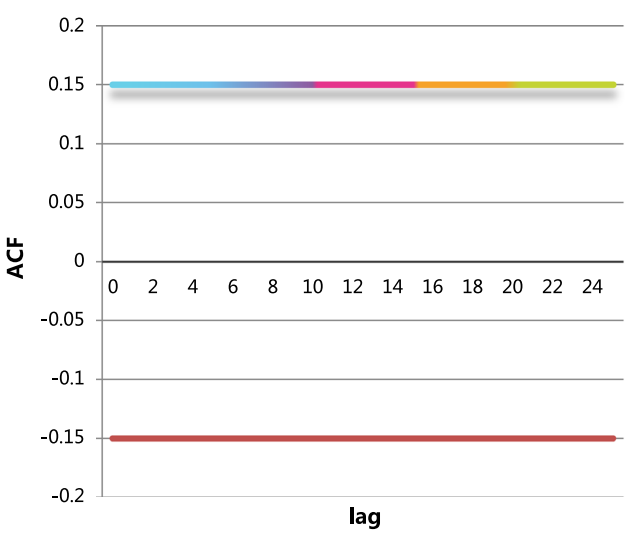

Fig. 7 ACF of residuals

the two rising trends after the dimensionality reduction were finally obtained as the input vector of the SVM.

(3) The above-mentioned SVM classification method is used to perform classification prediction experiments on the characteristic data of 200 sets of upward trend segments of agricultural product futures prices in 2016.

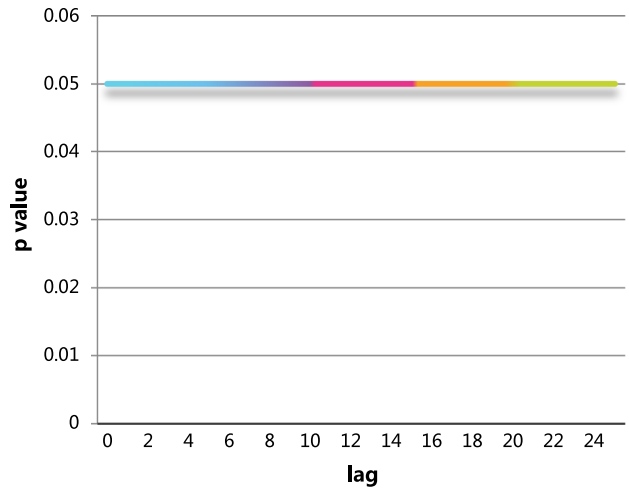

Fig. 8 *p* values for Ljung–Box statistic

(4) Evaluation of model prediction effect. In this study, the ROC curve was used to intuitively evaluate the prediction accuracy of the model, and the AUC test method was used to accurately represent the data. Moreover, the significance test was performed on the SVM classification model processed by the two data processing methods and the SVM classification model without data reduction.

(5) Model application research. The above-formed model with relatively good results was used to predict the new agricultural futures price data in 2017 and then compared and analyzed with the actual results:

7 Predictive experimental analysis

The data used in this study are derived from the futures historical data published on the official website of the Dalian Commodity Exchange. The main object of this paper is soybean futures. In addition, soybean futures prices, corn futures prices, etc., which have an effect on soybean futures prices, will also be studied to some extent.

Among the many indicators of the obtained data, the settlement price is selected for research to obtain the effect of the settlement price on the futures price. The settlement price is used to build a model and predict the futures price. The settlement price is the benchmark price for the day's trading margin and the day's profit and loss settlement for

Table 3 ARIMA (2,1,1) model prediction results

Date	Actual value	Predictive value	Lower confidence limit	Upper confidence limit	Prediction error	Prediction error rate (%)
2019.12.19	2919	2947.15172	2877.82532	3016.47711	1.03828	0.04
2019.12.20	2871	2947.95669	2836.41027	3059.50311	− 48.24669	− 1.68
2019.12.21	2833	2948.11021	2806.63446	3089.58596	− 86.78021	− 3.06
2019.12.22	2822	2948.14657	2780.1361	3116.15805	− 97.92657	− 3.47
2019.12.23	2803	2948.14859	2758.51806	3137.77811	− 117.11859	− 4.18
2019.12.26	2782	2948.15364	2738.01102	3158.29727	− 138.33364	− 4.97
2019.12.27	2797	2948.15061	2720.26936	3176.03085	− 123.18061	− 4.40
2019.12.28	2834	2948.15364	2703.0024	3193.30589	− 85.81364	− 3.03
2019.12.29	2827	2948.15061	2687.57768	3208.72354	− 92.88061	− 3.29
2019.12.30	2812	2948.15364	2672.39839	3223.90889	− 108.03364	− 3.84

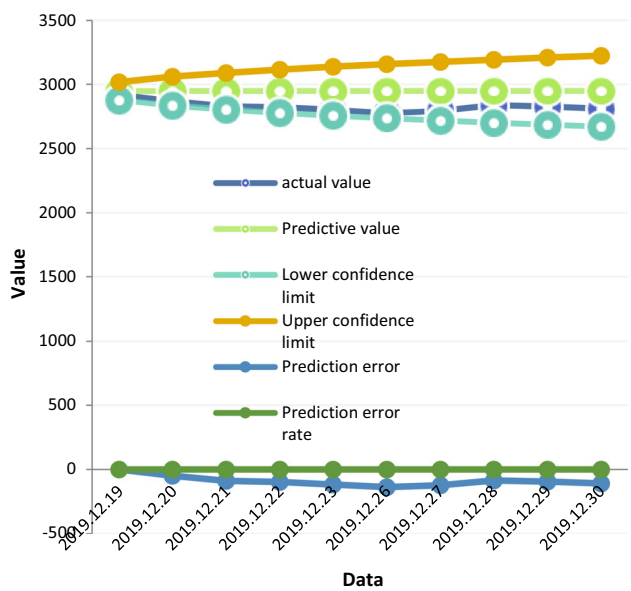

Fig. 9 ARIMA (2,1,1) model prediction statistics

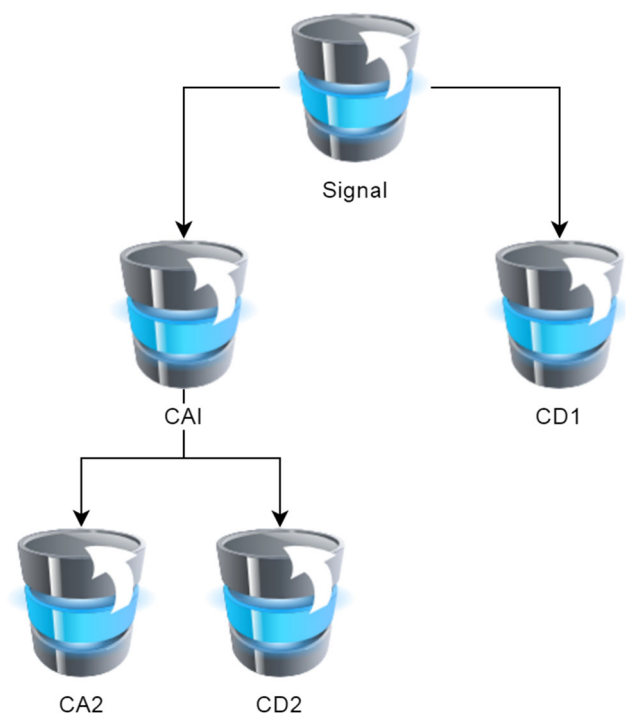

Fig. 10 Wavelet analysis decomposition diagram

open contracts after the day's trading. Moreover, the settlement price on the day refers to the weighted average price of the transaction price of a futures contract in the last hour according to the volume, which is a processed price, so it is more stable and less noisy. The main software used in this article is R. The timing of futures settlement prices is shown in Fig. 1.

Obviously, after the first order of difference, the sequence is stable, and the output shows that the P value is small enough to reject the null hypothesis that the sequence is not stable. Similarly, we can draw the timing diagram again to check its smoothness. The timing diagram after the difference is more stable and no longer shows a certain trend. The timing diagram after the difference is shown in Fig. 2.

In the third step, after the model is stable, the parameters in the ARIMA model can be determined, and the model can be ranked. According to the introduction of the previous theoretical knowledge, we need to obtain the autocorrelation map and partial autocorrelation map of the sequence after the difference. This work also needs to be completed by R. The output results are shown in Figs. 3 and 4.

The sequence autocorrelation graph after difference has a more obvious first-order truncation characteristic, so the value of the parameter q can be considered to be 1. The nature of the partial autocorrelation map after difference is

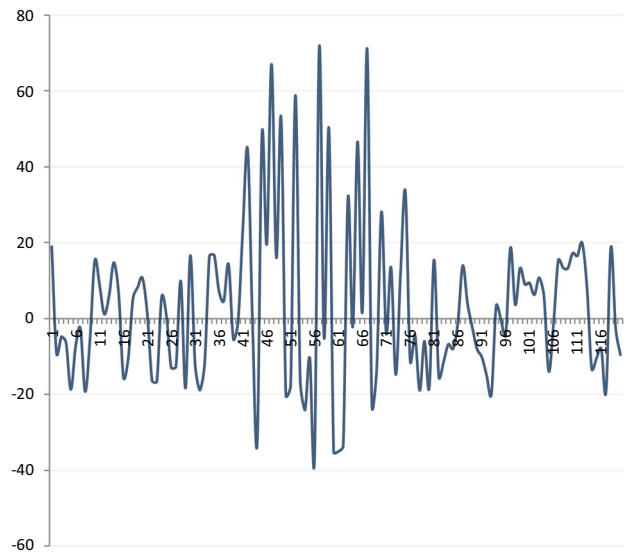

Fig. 11 Model-D1

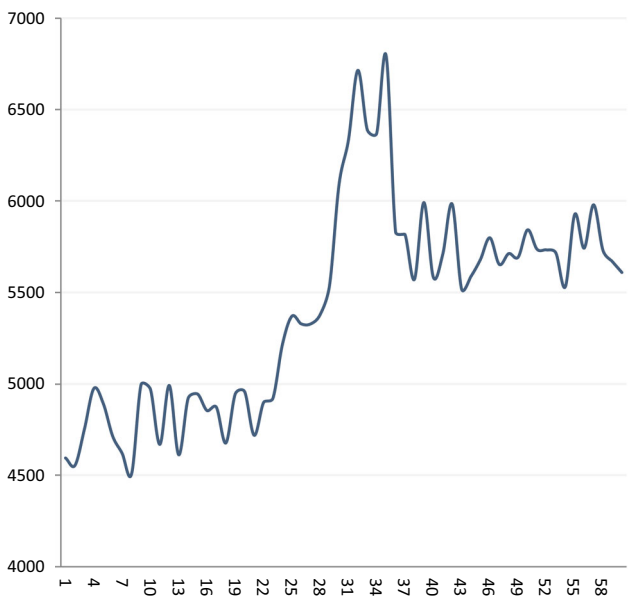

Fig. 12 Model-D2

not obvious enough, the correlation is small after the first order to the fourth order is delayed, and it shows a truncation after the fourth order is delayed. Therefore, we consider that the value of parameter p is 1, 2, 3, 4. Moreover, we use R software to build models ARIMA (0,1,1), ARIMA (1,1,1), ARIMA (2,1,1), ARIMA (3,1,1), ARIMA (4,1,1). Among them, it also includes the output result ARIMA (0,1,1) of the auto.arima () function for automatic ordering of the model in R. Based on the model selection criteria introduced in the theory section, the best ARIMA model is selected from the above five models. The AIC and model significance results of the five models are shown in Fig. 5 and Table 2.

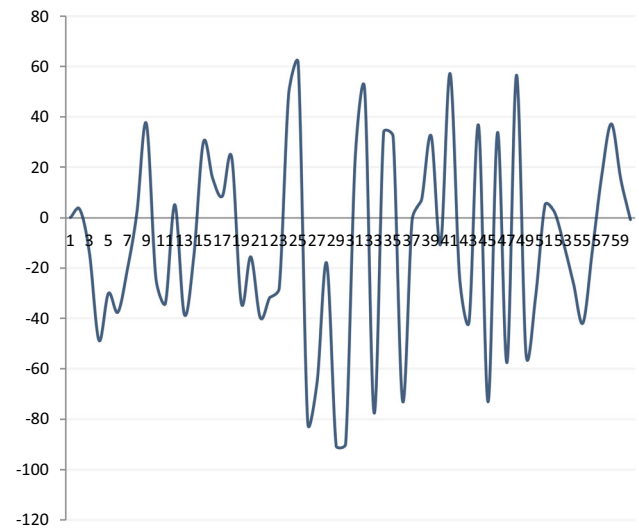

Fig. 13 Model-D3

Table 4 Model prediction results

Date	Actual value	Predictive value	Prediction error
2019.12.19	2948.19	2940.928403	7.261597
2019.12.20	2899.71	2928.170588	− 28.460588
2019.12.21	2861.33	2917.023521	− 55.693521
2019.12.22	2850.22	2922.957675	− 72.737675
2019.12.23	2831.03	2923.273199	− 92.243199
2019.12.26	2809.82	2929.550955	− 119.730955
2019.12.27	2824.97	2927.293908	− 102.323908
2019.12.28	2862.34	2914.998168	− 52.658168
2019.12.29	2855.27	2913.681532	− 58.411532
2019.12.30	2840.12	2920.156844	− 80.036844

From the perspective of model saliency, the latter two models are not significant and are not considered. According to the AIC criterion, it can be known that the AIC index of ARIMA (2,1,1) is the smallest. Therefore, we choose the model ARIMA (2,1,1) as the prediction model.

Another aspect of the good or bad test of the model is to test the residuals of the model to determine whether there is heteroscedasticity. The test results obtained using R software are shown in Figs. 6, 7, and 8. It can be seen from the output that the residuals fluctuate around 0, there is no trend, and the LB test has also passed, which indicates that there is no heteroscedasticity. The autocorrelation diagram shows that there is no autocorrelation between the residuals.

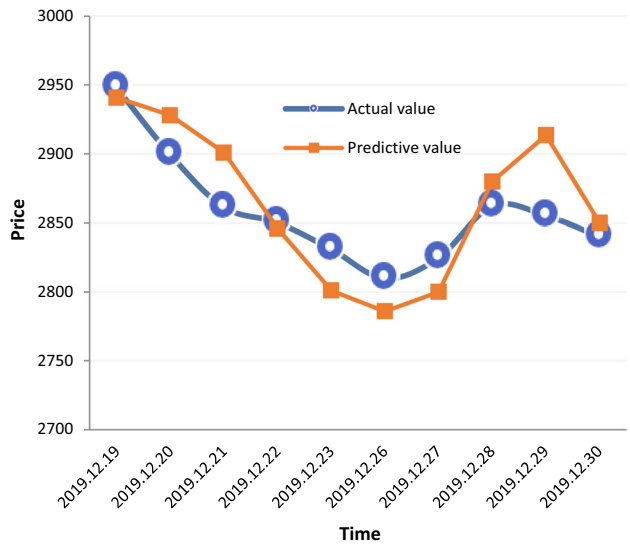

Fig. 14 Comparison of model predictions

After the model is established, the model is used to predict the test set. The 244 data are divided into the first 234 and the last 10, which are used as the training and test sets, respectively. Moreover, this study uses the model ARIMA (2,1,1) to obtain prediction results, including the predicted value and the 95% confidence interval. In this study, it is arranged in a table with the real value, and the prediction error, error rate, and cumulative error rate are calculated, as shown in Table 3 and Fig. 9.

As can be seen from Table 3 and Fig. 9, the prediction error rate of the 10 data in the test set is controlled within 5%. In particular, in the first three days of the forecast, the forecast error rate remained basically at 3%. Moreover, the error rate on the first day is basically negligible, and the magnitude of the absolute value of the cumulative relative error rate is acceptable. In general, the prediction effect of the model established is acceptable.

In order to be able to compare the prediction results of the model after adding the wavelet analysis with the prediction results of the ARIMA model, the 244 data of 2019 soybean meal futures are still divided into a test set and a training set. Among them, the first 234 data are used as the training set, and the last 10 data are used as the test set.

The data is decomposed using Python software. The decomposition process of the signal components is shown in Fig. 10.

The Python software performs signal decomposition on the data to obtain the signal components of each layer, as shown in Figs. 11, 12, and 13. From top to bottom are the high frequency of the first layer, the low frequency of the second layer, and the high frequency of the second layer.

From the fluctuation characteristics of each layer of components, it can be found that the fluctuation characteristics of the low-frequency sequence are most similar to the original sequence, and the high-frequency components show randomness and represents the fluctuation of a small part of the data of the original sequence, which is the part of market fluctuations that goes beyond smooth changes.

After obtaining the time series model based on wavelet analysis, we can use the established model to make predictions. The ARMA model established by each layer component is used for prediction, and then, the obtained prediction values are added to obtain the final prediction result. To facilitate comparison with the prediction results of the AEIMA model, we also put the true value and the predicted value in the same table, and calculated the prediction error, the prediction error rate, and the absolute value of the cumulative relative error. The results are shown in Table 4 and Fig. 14.

8 Conclusion

The research focus of this article is whether the machine learning model can be applied to the prediction and simulation of the commodity futures market, and whether the application is reasonable and superior. Starting from the analysis of the influencing factors of futures market prices, this article first clearly defines the relevant concepts of the futures market, systematically analyzes and introduces the relevant economic theoretical basis involved in futures price prediction, and briefly analyzes and elaborates the theoretical knowledge involved in futures price prediction models. Based on the analysis of the fundamental factors affecting the price of agricultural futures, the machine learning method is mainly used to predict the futures price. Moreover, the wavelet analysis method is used to smooth the data before establishing the ARMA model. The layered information after the signal decomposition is stable for each layer after the signal decomposition, so the ARMA model can be established directly. In addition, this study uses the established model to predict the settlement price of soybean meal futures. In the case of time series that are also univariate, the model is better than the simple ARIMA model.

Compliance with ethical standards

Conflict of interest The author has no competing interests.

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